

Homework 1

Problem 1: Vowel space extraction and vocal-tract-length normalization (50 points)

The zip file [hw1_data.zip](#) uploaded to Canvas contains a dataset of short, labeled .wav files (16 kHz, linear PCM, mono) of vowel productions from three speaker groups (men, women, and children), each producing 12 vowels.

(a) Formant extraction (15 points)

Extract F0, F1, F2, F3 and F4 for every token using [Parselmouth](#)'s Burg formant tracker, using a separate max-formant ceiling for each speaker group (men: 5,000 Hz; women: 5,500 Hz; children: 8,000 Hz). File [timedata.dat](#) gives the start, end, and steady-state times for the vowel in each recording; use the steady-state window, not the whole file. Rather than measuring at a single instant within that window, sample formants at several points across it (e.g., every 10 ms) and average, which is more robust than depending on where a single measurement happens to land in the window. Save your full extracted table (one row per token) as a CSV ([extracted_formants.csv](#)) since you will need it for the second problem; do not reproduce it in your report.

Select 3–4 example tokens spanning the three speaker groups, plot the waveform and spectrogram with the steady-state window and your extracted formant track overlaid, and use these to visually confirm the window is landing on the vowel and the tracker is following the right harmonics.

Then validate by comparing your extracted F1 and F2 against the ground-truth formant table in [vowdata.dat](#). Produce two scatter plots (extracted vs. ground-truth F_i) each with a $y = x$ reference line and points colored by speaker group. Report mean average error (MAE) and correlation between your extraction and the ground truth, separately for each speaker group, and indicate whether any group shows a systematic bias in its scatter plot (points consistently above/below the line) rather than just scattered noise.

(b) Group differences in vowel space (10 points)

Plot the vowel space (F1 vs. F2, inverted axes per lecture), color-coded by vowel, one scatterplot per speaker group. The three scatterplots should share the scale on the F1 and F2 axis so you can compare them. For every vowel, report the mean F1 and mean F2 across men, women, and children, and describe the direction and rough magnitude of the shift across groups using your actual numbers. What do you find?

(c) Vocal-tract-length estimation (10 pts)

For each speaker, estimate vocal tract length (VTL) using the [formant-dispersion method](#): model the vocal tract as a uniform tube, closed at the glottis and open at the lips, whose formants satisfy

$$F_n = (2n - 1) \times c / (4L)$$

Average F1–F4 across that speaker's steady-state measurements, compute the average spacing between consecutive formants ($\Delta F = \text{mean of } F_2 - F_1, F_3 - F_2, F_4 - F_3$), and estimate

$$L = c / (2 \times \Delta F), \text{ using } c \approx 35,000 \text{ cm/s}$$

Report the mean estimated VTL for each of the three speaker groups, alongside a boxplot of VTL by group, and check the group means against expected ranges (adult men ~17–18 cm, adult women ~14–15 cm, children shorter depending on age). Flag and investigate any individual speaker whose estimate falls well outside a plausible range for their group.

(d) Formant normalization based on VTL (15 points)

Using your per-speaker VTL estimates, normalize every token's F1 and F2 by the ratio of that speaker's VTL to the corpus-wide mean VTL, and re-plot the vowel space with normalized values in the same format as part (b). For each vowel, compute the group centroid (mean F1, F2) before and after normalization, and report the average pairwise centroid distance between groups (man vs. woman, man vs. child, woman vs. child) in both cases¹. Plot these pairwise centroid distances as a box plot, before vs. after normalization, grouped by pair. State the specific percentage reduction in average centroid distance after normalization and identify any vowel or group pair where a meaningful gap remains. What do you find?

Hints

Reading a wav file. `parselmouth.Sound(path)` reads the file directly and gives you everything downstream needs in one object:

- `sound.values` is a 2D array, one row per channel; for the mono files in this dataset, `sound.values[0]` gives you the 1D sample array.
- `sound.sampling_frequency` gives you the sample rate (16000 for this corpus).
- `sound.get_total_duration()` gives you the file length in seconds, useful for sanity-checking against `timedata.dat`'s start/end times.
- `sound.xs()` gives you a ready-made time axis matching `sound.values[0]`.

Plotting the waveform and spectrogram. A `parselmouth.Sound` object already gives you what you need for:

- **Waveform:** plot `sound.values[0]` against `sound.xs()`.
- **Spectrogram:** the simplest route is `ax.specgram(samples, Fs=sampling_frequency)` from matplotlib directly on the raw samples; no need to compute an STFT by hand for this diagnostic.
- **Overlaying the steady-state window:** `ax.axvspan(t_start, t_end, alpha=0.3)` shades a vertical band between two x-values.
- **Overlaying your formant track:** if you saved the per-sample-time formant values from your extraction, `ax.plot(times, f1_values)` on top of the spectrogram (matched y-axis, Hz) shows whether your tracked formant follows a visible harmonic band.

Problem 2: Vowel classification from acoustic features (50 points)

Using the same corpus and your Problem 1 extraction results, build k-NN classifiers that predict vowel identity from three different feature representations, and use them to investigate how well each representation generalizes to unseen speakers during training.

(a) Feature preparation (5 points)

Assemble three feature tables, each indexed by `file_id` and retaining a `talker_id` column; you will need it later for the speaker-level splits in (b)–(d):

- **F1/F2:** from your Problem 1 extraction (`extracted_formants.csv`)
- **F0/F1/F2:** the same, with F0 added

¹ For each pair (e.g., men vs. women), you compute the distance between the two centroids (e.g., male and female) for each vowel, then you compute the average centroid distance across the 12 vowels.

- **MFCCs:** 13-dim MFCCs, averaged across the same steady-state window used in Problem 1(a) (e.g., use *librosa.feature.mfcc*).

Save each feature table as a CSV.

(b) Speaker-independent 5-fold cross-validation (15 points)

Using only the men + women speakers, first set aside a separate subset of adult speakers (e.g. 20%) to select k , and exclude them entirely from the 5 CV folds used below; this keeps k -selection speakers separate from the speakers used for final reporting and prevents data leakage. On that held-out subset, try a small range of k values (e.g. 1, 3, 5, 7, 9, 11) and pick whichever gives the best accuracy. Report the k you chose; you'll use this same k for every feature set and every fold below.

With the remaining adult speakers, split by speakers (not by tokens) into 5 folds, i.e., every token from a given speaker must fall entirely within one fold, so no speaker is ever split across train and test. For each of the three feature sets in part (a), run the 5-fold CV using your selected k : in each fold, train on the other 4 folds' speakers and test on the held-out fold's speakers. Report the mean and standard deviation of test accuracy across the 5 folds, and a confusion matrix summed across all 5 folds' test predictions.

Plot the 5 per-fold accuracies as a box plot, one box per feature set. Then run a paired significance test (paired t-test, or Wilcoxon signed-rank) between each pair of feature sets, using the matched fold accuracies; the same 5 speaker splits underlie all three feature sets, so the folds are paired, not independent. Report the three pairwise p -values. NOTE: with 3 comparisons a naive $p < 0.05$ threshold inflates your false-positive rate, so please apply a Bonferroni correction, and state which feature-set differences are statistically significant. Separately, name the vowel pair(s) most confused for the best-performing feature set.

(c) Generalization to children: raw vs. VTL-normalized formants (15 points)

Using the 5 models you trained in (b) for F1/F2 (one per fold, never seeing children), evaluate each on the full set of **children's tokens**. Report the mean and standard deviation of accuracy across those 5 evaluations and compare directly to (b)'s mean CV accuracy for F1/F2; state the specific point drop. Then repeat using VTL-normalized F1/F2 from Problem 1(d): retrain 5 models on normalized adult folds, then evaluate each on normalized children's tokens.

Plot both sets of 5 accuracy values (raw adult vs. child, normalized adult vs. child) as box plots, and run a paired significance test between them (same 5 folds underlie both). Report the p -value and state whether VTL normalization's improvement in adult vs. child generalization is statistically significant.

(d) MFCCs and generalization (15 points)

Repeat the same adult-trained, children-tested evaluation using your 5 MFCC models from (b). Report the mean/std accuracy and the drop relative to (b)'s MFCC CV accuracy. Add MFCCs' 5 adult vs. child accuracy values to (c)'s box plot, and run a paired significance test between MFCCs' drop and raw F1/F2's drop from (c). Is one representation's generalization loss statistically distinguishable from the other's or not?

In 3–5 sentences, discuss why MFCCs might generalize differently than raw formants across speaker groups, and separately discuss whether F0/F1/F2's advantage over F1/F2 (if any, from (b)) survives the adult vs. child transfer in (c)/(d). Your discussion must cite specific mean accuracy numbers and p -values, not just describe the expected direction of the effect.

Deliverables

- A code archive (.zip or a link to a Git repository) containing all scripts/notebooks needed to reproduce every number and plot in Parts 1–5, plus a short README describing how to run it.
- A single report (PDF) containing every requested plot, table, and reported value from Parts 1–5, and a discussion of what those plots show.
- Do not submit the dataset, only your code and report.

Academic Integrity

This is individual work. You may discuss general concepts (e.g., “what does the vowel space tell us?”) with classmates or use documentation for NumPy/SciPy/Matplotlib/Praat, but all code and all reported numbers must come from your own runs. Do not use an AI coding assistant to generate solutions to any part of this assignment; you may use one to debug syntax errors in code you have already written. When in doubt, cite what help you received in a short “Acknowledgments” note at the end of your report. Disclosed help is never penalized; undisclosed copying is an integrity violation.